UNIVERSITY OF SANTO TOMAS
Faculty of Engineering
ELECTRONICS ENGINEERING DEPARTMENT
A.Y. 2026 – 2027

Undergraduate Thesis Conformity of Revisions

Thesis Title: **An Evaluation of Multi-Agent Reinforcement Learning for Dynamic Bus Scheduling Under Non-Ideal Conditions**

Group Number: B3

Conformity Revision Number: 1
Total number of recommendations: 22
Date of Submission: 08/29/2026

[X] Proposal Oral Examination   [ ] Final Oral Examination

Proponents		
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Research Technical Committee (RTC)			
Complied (C) / Not Complied (NC)	**RTC Members**	**Additional Comments** (May add additional sheets if necessary)	**Signature and Date**
	(RTC Member 1)		
	(RTC Member 2)		
	(RTC Member 3)		
	(Thesis Adviser)		

The detailed list of recommendations and corresponding revisions is found after this page.

Undergraduate Thesis Conformity of Revisions

Summary of Recommendations and Revisions				
No	**Recommendations**	**Description of Revision**	**Section No.**	**Page No.**
1	Include other figures/tables from the presentation that should also be in the manuscript.	Added the Route 801 corridor map (Figure 1.3) shown in the presentation, plus the parameter-summary and disturbance-coverage tables. Remaining slide content duplicated text already in the manuscript.	1.1	11
2	Expound on how traditional, non-AI scheduling systems perform under the conditions specified (bus bunching, severe weather, etc.).	Added an explicit account of how traditional strategies fail under the specified conditions — fixed timetables (no feedback), forward-headway (blind to a broken bus's backward gap), and even-headway (cannot anticipate heavy-tailed weather delays) — in §1.2.1, reinforced in the baseline-controllers section (§3.2.8).	1.2.1	13
3	Include study that considers severe weather conditions in your comparison.	Added Sun et al. (2025), an empirical study of bus dwell time under rainfall, to the ML/SARL disturbance-coverage table as the severe-weather (W) evidence.	1.2.2	16
4	Include study that considers severe weather conditions in your comparison.	Addressed together with the preceding severe-weather item: Sun et al. (2025) added as the severe-weather comparison study in the ML/SARL coverage table.	1.2.2	16
5	Consider adding a ML/SARL VSP table showing the disturbance column (S/T/W/B).	Added an ML/SARL disturbance-coverage table using the same D/S/T/W/B notation, situating the ML and SARL studies before the MARL comparison.	1.2.2	16
6	Some figures do not have citations (e.g., Figure comparing SARL and MARL).	Added source attributions to the concept figures: the SARL-vs-MARL figure now credits Gupta et al. (2017) and Buśoniu et al. (2008); the CTDE figure credits Lowe et al. (2017); the training-cycle figure credits Terry et al. (2021). Original diagrams are marked 'Authors' illustration.'	1.2.3	17
7	Explain the concepts in the SARL-vs-MARL figure, such as the bus states and actions	Added a paragraph explaining the figure's concepts — the per-bus state (position, forward/backward headways, onboard load, queue) and the holding/skip action — and how the centralized SARL panel and parameter-shared MARL panel differ.	1.2.3	17

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No	Recommendations	Description of Revision	Section No.	Page No.
8	In the MARL VSP table there is only 1 paper with breakdown (B); the presentation showed two. Update to accurate information.	Corrected the MARL comparison table: Shi et al. reclassified as demand + uncertain travel time (D,T) rather than breakdown; Cao et al. and Guedes & Borenstein added as adjacent breakdown-control evidence, reconciling the breakdown count with the presentation.	1.2.4	20
9	The research gap should include how you arrived at the column of sudden weather disturbance.	Added a paragraph to the Research Gap deriving the weather-disturbance class (W): the literature survey found no MARL bus study models heavy-tailed weather-induced delays, its operational relevance is established by rainfall's documented speed/capacity impact (§1.1), and it is parameterized via the validated lognormal travel-time form (Patil et al.).	2.2	24
10	Explain the justification why the minor roads leading to the corridor are no longer considered.	Added a justification in Scope and Limitations: feeder/minor roads are not simulated because agents observe only route-level headway, load, and queue states, and feeder-road effects already enter through the calibrated per-stop demand distributions.	2.5	27
11	Summarize the fixed and variable simulation parameters, including target values.	Added Table 3.2 summarizing all fixed, swept/variable, and derived simulation parameters, each with a target value or an explicit basis (verified-source or design value to be finalized in the implementation phase).	3.2.4	35
12	Update the manuscript with the proposed setup and discussion of dataset.	Added a dedicated dataset description: the CapMetro Rapid Route 801 APC dataset (Socrata asset im6q-3pc9, July--December 2021), its 47 fields, the cleaning/filtering rules, the 229,421-event direction-code-6 subset, and the NOAA weather join. Introduced in §1.1 and detailed in §3.2.5.	3.2.5	36
13	What are the contents of the dataset? Not described in the manuscript.	Added a description of the dataset's contents — the 47 APC fields and	3.2.5	37

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		the specific fields used (boardings, alightings, onboard load, dwell time, segment travel time/distance, coordinates, quality flags) — in the Data Processing section and Table 3.3.		
14	Provide mapping of dataset to proposed features of the study.	Added Table 3.3, mapping each verified CapMetro/NOAA field to the quantity derived from it and the role and limitation that quantity carries in the model.	3.2.5	37
15	Define each 'disturbance' explicitly, state dependencies, and distinguish stochastic demand from demand surge.	Added a 'Disturbance Classes and Independence' block defining D, S, T, W, and B explicitly, stating that the injected generators are sampled independently (no causal chain such as breakdown causing a surge), and distinguishing baseline stochastic demand (D) from demand surge (S).	3.2.6	38
16	Reference [10] is not quite new and simulates a different corridor. Will you adopt the same information or tune for the study corridor?	Clarified that the older rainfall study is used only as background. Its speed-reduction figures are not adopted; weather is calibrated from the route's own NOAA observations and a validated lognormal form (Patil et al.). The reference the panel flagged as [10] during the defense is now numbered [9] (Espino et al., 2018), since the references were reordered in this revision.	3.2.6	41
17	Explain in detail the different scenarios and how to simulate this data (traffic congestion, etc.).	Added an implementation-mechanics description to each disturbance generator (demand surge, traffic-speed, weather, breakdown) specifying when each random value is sampled and how it is applied during a simulated day.	3.2.6	39
18	Include the details on the metrics and description of features.	Added formal definitions of the three response metrics (mean passenger waiting time, mean travel time, headway coefficient of variation) and Table 3.6 describing the agent observation features and their real-world versus simulation sources.	3.2.9	53
19	Can you describe what a successful performance will look like?	Added explicit Stage A and Stage B acceptance criteria defining a	3.2.10	54

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		successful controller (parity with or improvement over Even Headway on waiting time, and a significant reduction in headway coefficient of variation versus No Control).		
20	When putting figures and tables, they should also be called and discussed in the paragraphs.	Every figure and table now has an explicit in-text callout and is discussed in the surrounding paragraphs; applied throughout Chapters 1 and 3.		
21	Consider using 1.5 line spacing for easier readability.	Applied 1.5 line spacing throughout the manuscript.		
22	Consider putting line numbers for non-final manuscript versions.	Enabled line numbers throughout the manuscript for this non-final version.		
	Nothing Follows			
Summary of Recommendations and Revisions				

Notes:

1. Make sure to fill out all details carefully. Inaccurate details may cause delays in getting approval of your proposal revisions.
2. The latest manuscript must accompany this form. Enable line numbers in the manuscript for easy referencing. The group is highly encouraged to highlight the revisions done on the pdf copy of the manuscript so that RTC members can easily find the revision.
3. In the “Summary of Recommendations and Revisions” table
    - Add “Nothing follows” as last row entry
    - Take note that the page number should be based on the manuscript page number and not based on the PDF page number.
4. For submission:
    - This form must be signed by all group members and the adviser.
    - Submit the PDF copy on the course site.
    - The PDF copy will then be forwarded to the RTC members by course instructors for their review.
    - If RTC members find that recommendations are not addressed (e.g., not complied with), they may ask to resubmit. The group will then repeat the process and prepare an updated version.